ZentroMart VoiceOps

Capstone Project Report

Synthetic customer support with retrieval, guarded tools, and Gnani voice adapters

Prepared	Dataset reference date	Selected LLM
22 October 2026	20 October 2026	Qwen3 4B (local Ollama)

Executive summary

This capstone builds a customer-support assistant in four stages: a prompt-only local Qwen3 4B baseline, retrieval over the supplied synthetic company knowledge base using local embeddings and a vector index, a multi-turn agent with verified order lookup and guarded ticket creation, and Gnani speech-to-text/text-to-speech adapters. The hybrid retriever found every expected source for 47/47 answerable practice questions at top five. The final RAG result is 52/52, combining a complete 48/52 run with a focused replay of four corrected edge cases (4/4). The final multi-turn result is 96/96, combining a complete 92/96 run with a focused replay of four corrected scenarios (17/17 turns). Synthetic single-turn and one-file batch voice checks passed end to end with Gnani STT/TTS and local Qwen. A user-recorded microphone sample, multi-file voice order-verification dialogue, and the required short demo video remain to be completed.

Problem and goals

Customers ask general policy questions and questions about specific orders. The assistant must answer from company evidence, verify identity before disclosing order details, avoid leaking personal data, and create a human handover only for qualifying cases. The final voice flow must transcribe audio, run the support flow, synthesize a spoken reply, and save evidence and tool-call records for each recording.

Implementation

Stage 1 - Prompt-only baseline

The local Qwen3 4B model answers without retrieval. This creates a simple baseline and demonstrates the limits of general model knowledge. Ollama serves the model locally; no hosted LLM API key is required.

Stage 2 - Retrieval-augmented generation

A local nomic-embed-text model embeds policy sections and approved catalog/pincode tables. A persistent vector cache stores embeddings outside the submission package. Retrieval fuses cosine similarity with BM25, exact SKU matching, multilingual search variants, and policy routes for safety-critical topics. Retrieved source IDs are included in the answer flow. Business-day refund dates use the policy duration and holiday calendar; return deadlines use calendar-day arithmetic.

Stage 3 - Multi-turn tools

The agent stores conversation state in memory and uses the dataset local mock tools for order lookup and ticket creation. Specific order details require an order ID plus the phone last four digits or delivery pincode. Verification proof and direct contact fields are removed from model context and ticket summaries. Ticket creation is gated on an explicit human request or policy escalation. These mock tools are educational, not production services.

Stage 4 - Voice

Gnani STT and TTS use the official gnani-vachana Python SDK. The batch runner accepts mono 16 kHz, 16-bit PCM WAV and saves transcript, assistant text, synthesized WAV, tool-call record, and retrieved evidence IDs. The synthetic single-turn and one-file batch paths passed end to end with Gnani STT/TTS and local Qwen. The batch runner also supports one session across sorted audio turns with --conversation. A user recording, multi-file voice order-verification dialogue, and the short demo video remain to be completed.

Evaluation results

Stage	Evaluation	Result	Interpretation
Prompt-only	52 practice questions	8/52	Literal fact check; no knowledge base; 49/52 escalated conservatively.
Retrieval	47 answerable; top-k 5	47/47	All expected evidence found for the answerable subset.
RAG	52 practice questions	52/52	100% decision accuracy and gold-source recall; final four edge cases confirmed by focused replay.
Multi-turn	96 supplied turns	96/96	Full run 92/96 plus focused replay 17/17 across four corrected scenarios.
Gnani voice	One synthetic input; single + batch	Passed	Official SDK STT/TTS with local Qwen reply; real recording and demo remain pending.

Interpretation limits. The practice set is synthetic and small, and results do not guarantee performance on unseen evaluation prompts. The RAG result combines the complete 52-question hybrid-retriever run (48/52) with focused replay of four corrected cases (4/4). The tool-agent result combines a complete 96-turn run (92/96) with focused replay of four corrected scenarios (17/17).

Run instructions

Use Python 3.10 or newer, install Ollama with qwen3:4b and nomic-embed-text, install requirements.txt, and extract the supplied dataset to work/dataset/Capstone_Project_Dataset_FTTT_2026. Voice features also require GNANI_API_KEY in a private .env file. Never commit .env.

```
python run_voice_batch.py path/to/audio_folder --language en-IN --voice-language auto
```

Voice inputs must be mono 16 kHz, 16-bit PCM WAV. The supplied dataset contains no audio recordings. Full baseline, retrieval, RAG, and tool-agent commands are in the project README.

Submission status and next steps

The source package is prepared without the dataset, embedding cache, virtual environment, generated ticket file, or secrets. A public GitHub repository is available; a deployed demo and required demo video (maximum two minutes) remain outstanding. Record a user voice interaction, demonstrate multi-turn voice order verification, and produce a short demo showing a document-grounded question and follow-up plus a handover or safe refusal.

Sources: supplied VoiceOps 2026 Capstone Brief and synthetic FTTT dataset (README, policy files, practice data, tables, and mock tools); Gnani official Python SDK (gnani-vachana) and API documentation at docs.gnani.ai/api; Ollama local model runtime with Qwen3 4B and nomic-embed-text.